Research questions for an incomplete crowd

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June 2026

1 Summary

What are the points I am trying to outline [1]

- Many of the questions out of the 27 I argue IIASA is simply not prepared to do. There is too much focus on research who are leaders in hard quantitative methods, especially around modelling. However I argue that many of the question require the input of qualitative researchers with a strong focus on theory. There is fundamentally a lack of researchers at IIASA who specialize in generating theory. Without this the outlined questions cannot be answered. We can only answer with what theory already exists and too many of the questions are too forward looking to not require deep and innovative new theory.

- As a proposed solution I believe the institute must either learn from interdisciplinary institutions such as the Santa Fe Institute or the Vienna Complexity Hub, or should reconsider the scope of proposed research direction to account for the composition of its researchers, or lastly have interactions with a broader range of researchers as part of its explicitly future research plans. IIASA also benefits from an much stronger international background of researchers than these other institutes from which it can pull new visions of a political future.

- IIASA as an explicitly international and diplomatic research institute cannot, and potentially should not answer questions of political economy. Put bluntly the financial crisis that IIASA faces means it cannot risk antagonisation of its NMOs and RMOs. I again show my lack of knowledge but I assume a question such as "What would a political economy for a post-industrial world look like?" would not have gone down particularly well with either the Soviet (nominally socialist) or US (explicitly free-market capitalist) NMOs in the 1970s and 80s given the implication of a transition away from their respective economic models. The questioning of the status quo is most likely going to lead those nations that are already under the sway of strong anti-science and xenophobic sentiment such as the US to retreat even further from the organisation funding wise. The historical purpose and funding mechanisms of the institute simply do not allow it the freedom to question the inequality that maintains the imperial core rich (US, EU, Japan etc) and the periphery poor (global south). A solution to this may simply be a move away from nation based funding or a focus on greater funding from Global South nations that have more to gain from a just post-capitalist transition.

- Questions which I think face a lack of expert resources at IIASA are:

[1] As will shortly be made obvious in the loose and often dubious quality of the writing of this letter, it was written without the aid of AI.

  - What would a political economy for a post-industrial world look like?
  - How will disruptive innovations reshape our societies and how can we ensure that the benefits are shared equitably to serve the common good?
  - What is a fully regenerative economy and what kind of prosperity can it sustain?
  - Can we quantify and eventually close the "regenerativity gap"?
  - What is human well-being all about in the Anthropocene?
  - Which governance and decision-making systems can enable societies to navigate polycrisis and adapt to the next crisis that has no name?

Further Points:

- A lot of these are also just bad research questions that can be answered yes or no, and not well structured to be specific enough to give useful or open enough that they can be dismissed off hand.

- I am fully aware of my own naivete at having worked at IIASA for a grand total of a couple months but sometimes an outside opinion can help spur a discussion in the right discussion. This argument then is based on my skewed, limited experience of IIASA's research and that of quantitative researchers as whole.

What I offer:

- My own (limited) interest, outside of modelling, is in the area of economics history and transitions. I use this as an example of where I believe the question being proposed are interesting but that IIASA holds half the answer to the question. I argue that researchers from the traditionally soft sciences must be integrated in order to tackle these questions. Specifically I tackle question 2: "What would a political economy for a post-industrial world look like"?

2 What would a political economy for a post-industrial world look like?

I begin with restating that I do not think that this is a bad question but merely that the composition of the institute by itself cannot adequately tackle it. I now outline what I as the issues with this questions, how it might be tackled and what the contribution of quantitative researchers based at IIASA might be?

2.1 Just because production is hidden, does not mean it is not happening

I first contest the concept of a post-industrial world. My understanding of the question, in the context of the growing role of AI in discourse, the panel at the interaction festival and the other 26 questions, is that AI will somehow replace human labour to the extent that it is no longer required. I will attempt to make a (bad) Marxist/labour-theory-of-value argument to demonstrate the infeasibility of this. The key argument here is that the political economy of a system does not experience a fundamental bifurcation or break unless the dynamics that affect the replication of classes within that system change. Our current form of production or societal organisation can be seen as an emergent property of a fundamental interaction that a worker sells their time to an employer in order to reproduce themselves through consumption of food or housing etc, as they do not have access to the means by which to reproduce themselves as they do not own fields/vegetable gardens or houses. I argue that AI will not break this dynamic unless under a specific set of circumstances, outlined below.

I consider three scenarios withe ever greater technological improvement: 1. Improved agentic AI but limited physical labour replacement 2. Wide spread physical labour through AI enhanced robotics 3. Complete replacement of human physical labour.

2.1.1 Just another calculator, car or computer

Assuming that thanks to agentic AI the productivity of office labourers continues to increase then I argue that this is a substitution of labour similar to that of the replacement of traditional "computers" by calculators or the replacement of secretary pools by personal computers. The effect of this replacement was heterogenous in both race, class and gender, for example both computer and and secretary pools were predominately women, but the effect is that a new group of workers enter a pool of reserve labour (unemployment) until they can retrain to find other new labour forms (often under worse conditions due to breaking of institutions such as unions that formed in these now outdate jobs). Crucially though the effect is cyclical and consistent throughout the 500 year old history of capitalism: A new technology increases labour productivity, leading to the displacement of labour with capital, this labour then finds employment in another area. The initial increase in profit rate of that sector or firm decreases over time as competitors are able to replicated the technology and things slowly reach an equilibrium as technology improvement slows down. A key dynamic is that the labour required for the reproduction of a worker may decrease over time with the increasing of productivity in sectors producing necessary goods such as food or travel. An example of this is the collapse of the share of labour that is dedicated to agriculture in the global north due to both mechanisation and greater importing of food.

Another dynamic here is the displacement of labour to other countries with lower cost (due to a lower cost of worker reproduction) this phenomena is exemplified by the long trend of China as the workshop of the world in the 21st Century. The point I raise here is that a "post-industrial" political economy may simply be one in which labour is displaced to marginalized communities within the global north or "hidden" in global south (A good example of this is the large scale labelling of data sets, or the use of VR workers for robots). Essentially what we have already in the 21st Century where "Made in Germany" cars are mostly made else where with 54% of working time occurring upstream (Pérez-Sánchez and Aguilar-Hernandez, 2026).

Fundamentally as long as we consider the system as a global one and not limited to the internal dynamics of global north nations then we can see that under limited labour displacement from AI the industrial labour required to extract, construct and supply the building blocks of global north society will still be driven by human labour, but that labour will not be seen by the global north consumer, as has been the case since the advent of globalisation.

2.1.2 Terminiator-lite

The second scenario I consider is one in which there is a rapid improvement of robotics and some physical labour is automated, but not all. Essentially this is a continuation of the previous scenario simply the scope of the analysis must be expanded. The cost of reproduction of a worker falls furthers as the human labour required in the production of necessary goods falls further. Under competition this leads to further falls in the prices of necessary goods, mass transitional unemployment, civil unrest etc but the fundamental reproduction of the working class does not change, they must still sell their labour to work in jobs that have not been replaced by agentic/robotic AI, such as care work, teaching, research or creative arts. A key limiting factor here will be the cost of replacing human capital with AI capital, as the labour time required for human reproduction falls so will the cost, whist the cost in the case of human labour for the expansion of AI may increase to not match this decrease. Human labour capacity may still be cheaper limiting the penetration of AI into labour markets. This scenario is a rapid race to the bottom where everyone loses, but, as usual, the poor lose faster.

However I think it is important to note that at this stage the falling usefulness of labour means that because of obscene wealth concentration, the capitalist may not have further interest in more automatization due to it simply reducing the size of their potential market. If they have no one to sell to then they cannot profit, hence the profit rate falls and capital flees to another sector, perhaps care work.

2.1.3 i-robot

In the third scenario I consider, what if an agentic AI can replace current human physical industrial and agricultural labour. This would mean a closed cycle from extraction of materials, to transport, to construction of steel, to the building of factories and energy systems. What then will happen to the working class? To answer this we must look again at the cost of reproducing the class. In this scenario, the cost of human labour reproduction now purely comes from the labour associated with the few things that cannot be automated, such as creative and care work. This incredibly low cost means that the value of human labour plummets further. A key bifurcation would be whether the spread of automation is complete, or whether pockets exist where the cost of human labour is still lower than that of capital due to frictions. Human labour may become a sign of opulence and luxury, a throw back to a simpler more romanticised, artisanal, time (that never existed but sells well in the sweltering heat of 2040).

An utopian might see this as an opportunity for some sort of "fully automated luxury communism" (Bastani, 2019) after a quick and fantasy-like revolution. An optimist would argue that given the lack of labour the working class might have time to organise and fight back to seize control of the automated means of production as they fight the clock against mass starvation due to a lack of ways to generate income. The pessimist might say that the ruling class no longer has a need for the working class in its current size, they have lost all ability to consume given that there is no labour for them to perform. Thus to the ruling class they are simply a physical threat, the solution to remove this class threat is gruesome to say the least.

The wage system is a fragile peace treaty[2] that prevents the poor from eating the rich. Therefore, in this future there is no future, there is no post-industrial political economy because there will be no polis, no society.

2.2 To go forward, you must go back

Having painted a rather dim, sad view of the future that I do not think qualifies as post-industrial political economy I now turn to argue what can a model contribute to this debate? I will attempt to draw parallels to how a feudal IIASA might have contributed to establishing how a post-feudal or post-agricultural society might look like at the cusp of the transition towards agricultural capitalism around 1400-1600. This argument and model is based on the work of the Political Marxist and Historian Ellen Meiksins Wood as outlined in "The Origin of Capitalism" (Wood, 2002), and the work of Robert Brenner (Brenner, 1976). I will first outline the theory of transition from Feudalism to Capitalism and then outline how IIASA's expertise might be applied to this question, then at last I outline how this relates to how with the aid of political theory a new political economy might be modelled by IIASA.

2.3 As usual with Marxists, everyone is wrong but us, even the other Marxists

During the 1970s the Brenner Debate produced an enhanced theory of transition from feudalism to capitalism based in rural 17^{th} Century England. The key wedge that Political Marxists introduced was that past historians were assuming an a priori existence of an embryonic capitalism within feudal society, thus rendering the very idea of a transition useless.

Brenner's theory sought internal conflicts within feudalism, as opposed to an external impetus, which might produce its collapse[3]. Wood offers an analysis that " it [was] a matter of lords and peasants, in certain specific conditions peculiar to England, involuntarily setting in train a capitalist dynamic while acting in class conflict with each other to reproduce themselves as they were" (Wood, 2002, p.52). This focus on the class conflict nature of the transition is what distinguished Brenner and Wood's theory from other contemporary Political Marxists. Brenner identifies the internal logic of feudalism that inhibits surplus reinvestment. For the feudal lord it was often easier to find new avenues

[2]In the most deeply ironic of fashions, this specific framing was suggested by Gemini.

[3]This and next paragraph ripped from my masters thesis.

to "squeeze" (Brenner, 1976, p.74) out more absolute surplus value from their subjects instead of increasing their labour efficiency. Due to the fact that a lord's power was directly correlated to size of the population under his control this lead to a focus on labour immobility, and maximising the ease with which he could extract surplus value from his peasants by non-economic means. Resulting in restrictions on land mobility and "unfree peasants [unable] to convey their land to other peasants without the lord's permission" (Brenner, 1976, p.50).

A key tenant of this school of thought is the role of "improvement" as both a motor in the divorce between peasants and their means of subsistence, and a consequence of newly created market imperatives both landlords and tenants were now subject too. The emergence of a market dependence of peasant to access land meant that there was now an imperative to be more competitive in agricultural productivity spurring technological organisation and kick-starting proletarianisation and the further enhancement of enclosure.

2.4 The current ability of IIASA is to model theory

In this case study the key dynamic to be modelled is how interactions between classes result in the emergence of a new means of class reproduction, and thus a new arrangement of power. Such a model would focus on how changing technological, ecological and economic conditions allow for conflicts to emerge between peasants and landlords. This conflict might be represented using an Agent-Based or System Dynamics model in addition to a model of land use, all of which IIASA's has ample expertise in.

2.5 The future ability of IIASA can be to generate and model theory

Our feudal IIASA requires not modellers who are methods experts but those who can propose what a new world might look like, predicting how the nascent conflict of classes in England may resolve themselves. When working in collaboration, with theory and numerical model together IIASA will have the ability to answer such demanding question as "How did feudalism transition to capitalism" and the more pressing "What might a post capitalist, maybe post-industrial, political economy look like?".

However as outlined at the start of this letter, our feudal IIASA is faced by the uncomfortable fact that this new political economy is contrary to the feudal economy where the landed aristocracy dominate. This shift from feudalism is the start of the long decline of the aristocracy and the emergence of a new super rich, the industrialists (Alfani, 2023). Therefore why would they fund such a piece of work which so evidently undermines their class.

References

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A IIASA Research questions

1. How can systems analysis reconcile analytical reasoning, machine algorithmics, and human intuition into a coherent poly-rationality?
2. What would a political economy for the post-industrial world look like?
3. How will disruptive innovations reshape our societies and how can we ensure that the benefits are shared equitably to serve the common good?
4. What is the purpose of education and communication in the AI epoch and how can they strengthen human agency?
5. Is there a critical climate overshoot limit that we will not be able to recover from?
6. How can we bend the global temperature curve back into the safe operating space for humanity?
7. What is the realistic ceiling on drawing carbon back out of the atmosphere - and can we do it with societal benefit rather than at societal cost?
8. How can humanity and the Earth survive the overshoot episode?
9. Is the transformation of the built environment the crucial factor in the climate mitigation-adaptation equation?
10. Will the fusion of nature-based solutions with hi-tech save the world?
11. Do we need to reimagine the notion of ”development”?
12. What is a fully regenerative economy and what kind of prosperity can it sustain?
13. Can we quantify and eventually close the ”regenerativity gap”?
14. How can complexity be harnessed for reconciling competing demands on land, water, and ecosystems?
15. How much of future economic output could come from functioning ecosystems?
16. Does the clean-energy transition trade old fossil dependencies for new mineral ones - and can it be navigated without a new geopolitics of scarcity?
17. Can demand-side change accelerate the transition and build resilience in a volatile world?
18. How little energy and material does a decent life require - and can the transition deliver it to all?

19. What is human well-being all about in the Anthropocene?
20. Which governance and decision-making systems can enable societies to navigate polycrisis and adapt to the next crisis that has no name?
21. How can systems analysis identify and leverage transferable patterns across social, economic, and ecological systems to better understand complexity, anticipate disruption, and support transformative change?
22. Can societies remain governable when demographic majorities become minorities of power?
23. Can health become the organizing principle of resilient societies, rather than the repair system of societies in crisis?
24. Can mobility become a planetary adaptation system rather than a political failure?
25. How to accomplish science diplomacy in a multipolar, innovation-disrupted world?
26. Can IIASA attract and train the vanguard of future decision-makers?
27. Is the human technical intelligence alone in our galaxy, and if so, why?